Expert Systems with Applications 42 (2015) 7664–7675

Contents lists available at ScienceDirect

Expert Systems with Applications

journal homepage: www.elsevier.com/locate/eswa

Mixed dissimilarity measure for piecewise linear approximation based time series applications

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ARTICLE INFO

Article history:
Available online 10 June 2015

Keywords:
Dynamic time warping
Local distance
Piecewise linear approximation
Similarity
Constraining

ABSTRACT

In recent years, expert systems built around time series-based methods have been enthusiastically adopted in engineering applications, thanks to their ease of use and effectiveness. This effectiveness depends on how precisely the raw data can be approximated and how precisely these approximations can be compared. When performance of a time series-based system needs to be improved, it is desirable to consider other time series representations and comparison methods. The approximation, however, is often generated by a non-replaceable element and eventually the only way to find a more advanced comparison method is either by creating a new dissimilarity measure or by improving the existing one further.

In this paper, it is shown how a mixture of different comparison methods can be utilized to improve the effectiveness of a system without modifying the time series representation itself. For this purpose, a novel, mixed comparison method is presented for the widely used piecewise linear approximation (PLA), called mixed dissimilarity measure for PLA (MDPLA). It combines one of the most popular dissimilarity measure that utilizes the means of PLA segments and the authors' previously presented approach that replaces the mean of a segment with its slope.

On the basis of empirical studies three advantages of such combined dissimilarity measures are presented. First, it is shown that the mixture ensures that MDPLA outperforms the most popular dissimilarity measures created for PLA segments. Moreover, in many cases, MDPLA provides results that makes the application of dynamic time warping (DTW) unnecessary, yielding improvement not only in accuracy but also in speed. Finally, it is demonstrated that a mixed measure, such as MDPLA, shortens the warping path of DTW and thus helps to avoid pathological warpings, i.e. the unwanted alignments of DTW. This way, DTW can be applied without penalizing or constraining the warping path itself while the chance of the unwanted alignments are significantly lowered.

1. Introduction

In the last decades, one of the most important trend of expert systems has been the emerging interest in data-driven approaches. The databases are not seen as the simple input of a hard-coded decision system anymore, instead, they are also actively used to improve the system itself by utilizing data mining.

The availability of easily accessible computational resources and storage capacities has made it possible to apply classical data mining techniques on time series data. Such methods have rapidly gained attention in engineering applications, including process engineering (Singhal & Seborg, 2005), medicine (Tormene, Giorgino, Quaglini, & Stefanelli, 2009), bioinformatics (Aach & Church, 2001), chemistry (Abonyi, Feil, Németh, & Árva, 2005), finance (Rada, 2008), biometrics (Gavrila & Davis, 1995; Kholmatov & Yanikoglu, 2004; Vajna, 2000) and even tornado prediction (McGovern, Rosendahl, Brown, & Droegemeier, 2011) and sewer flow monitoring (Dürrenmatt, Giudice, & Rieckermann, 2013), because of the effective and efficient methods available for solving time series-related problems. But these methods are not limited exclusively to temporal data. Every task that can be transformed to a sequence-related problem is also a candidate, such as motion detection (e.g. gun draw detection (Ratanamahatana & Keogh, 2004)), handwritten word recognition (Rath & Manmatha, 2003), off-line signature verification (Piyush Shanker & Rajagopalan, 2007), detection of similar shapes in a large image dataset (Xi, Keogh, Wei, & Mafra-Neto, 2007), etc.

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http://dx.doi.org/10.1016/j.eswa.2015.06.003
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The increasing use of time series data has initiated a great deal of research and development attempts in the filed of data mining (Fu, 2011) and thus a continuously growing knowledge base has been built up to utilize time series data even better. Two things are, however, common in every time series-based data mining task: a representation of the temporal data has to be selected and a dissimilarity measure has to be defined between the represented values. The representation thus not only determine the tightness of the approximation but also the dissimilarity measure to use and, in the end, the result itself. Motivated by this fact, large number of time series representations — alongside the matching dissimilarity measures — have been introduced by the time series data mining community: episode segmentation (Cheung & Stephanopoulos, 1990), discrete Fourier transform and discrete wavelet transform-based approaches (Agrawal, Faloutsos, & Swami, 1993; Chan & Fu, 1999), Chebyshev polynomials (Cai & Ng, 2004), symbolic aggregate approximation (Lin, Keogh, Lonardi, & Chiu, 2003), perceptually important points (Fu, Chung, Luk, & Ng, 2008), derivative time series segment approximation (Gullo, Ponti, Tagarelli, & Greco, 2009), implicit polynomial curve-based approximation (Wu & Yang, 2013), etc. All proposed to compete and replace classical time series representations such as piecewise linear approximation (PLA) that has been used — and still preferred — in several engineering applications thanks to its intuitiveness and ease of visualization. Data acquisition systems also echoed this trend and in most systems different approximations of the raw data can be selected allowing to choose the one, which suits most the performed task.

In many cases, however, the representation cannot be changed as it is provided by an unchangeable hardware device or software component. Cars developed according to the AUTOSAR concept (www.autosar.org, 2003) are good examples of that. AUTOSAR is based on the assumption that elements (sensors, processing units, application logic, etc.) of a given automotive functionality made by different suppliers can be easily combined, if the interfaces between the hardware/software elements are strictly defined. It is not an uncommon situation when an analog sensor (the sensing element) and its control unit are provided by two different suppliers. Similarly, the basic software, which operates the control unit, and the application logic can also be created by separate companies. In such a complex and interdependent environment, an analog sensor is often coupled with an application-specific integrated circuit (ASIC) and/or a software library to provide — a usually highly simplified — digital representation of the analog signal. This representation is then forwarded by the control unit through the basic software to the application logic. As an example, Fig. 1 shows the analogue (raw) signal of an automotive ultrasonic sensor and its PLA representation, which can be used by the application logic. Due to the already given representation, the only possibility for the application logic provider to gain more accuracy is not to modify the representation itself but the dissimilarity measure.

Although the creation of mixed dissimilarity measure for PLA (MDPLA), presented in this paper, was motivated by a similar situation —, i.e. PLA segmentation could not be changed —, the development was also driven by the need of minimization of pathological alignments often created by dynamic time warping (DTW[2]).

2. Alignment problem of dynamic time warping

The unwanted alignments are created when the feature considered by the local dissimilarity measure of DTW is similar between a relatively small section of one time series and a much larger section of another time series. This phenomenon can usually be seen when DTW *fail to find obvious, natural alignments in two sequences simply because a feature (e.g., peak, valley, inflection point, plateau etc.) in one sequence is slightly higher or lower than its corresponding feature in the other sequence* (Keogh & Pazzani, 2001).

To avoid such unwanted alignments, global constraints, which limit the warping path how far it can stray from the diagonal, have been introduced by Itakura (1975) and Sakoe and Chiba (1978) (see Fig. 2). According to their experience, *time-axis fluctuation in usual cases never causes too excessive timing difference*, and thus pairing points far away in time would eventually degrade the results.

Limiting the warping path has another advantage, it speeds up the calculation of DTW by a constant factor. As global constraints limit the search space in the warping matrix, local dissimilarity is only necessary to be computed inside the constrained area.

Due to the expected better results and the speed up, most practitioners dealing with DTW utilizing of some form of global constraints independently of whether DTW was used for off-line handwritten signature verification (Güler & Meghdadi, 2008) or for sewer flow monitoring (Dürrenmatt et al., 2013). However, selecting the optimal global constraint — deciding not only which constraint should be used, but whether any constraint should be used at all — is not obvious. When DTW has been used tasks other than speech recognition, *researchers have used a Sakoe-Chiba band with a 10% width for the global constraint* [...] *result of historical inertia inherited from the speech processing community, rather than some remarkable property of this particular constraint* (Ratanamahatana & Keogh, 2005). The width of the Sakoe-Chiba band was optimized later in several applications, e.g., by Long, Fonseca, Foussier, Haakma, and Aarts (2014); however, such an approach can provide a global optimum only and leaves space for local pathological warpings. Thus, to have a better control of the warping path — and to improve DTW results —, different approaches have been utilized.

Ratanamahatana and Keogh (2004) presented the R–K band, which uses a heuristic search algorithm that automatically learns the constraint from the data and locally shrinks the Sakoe-Chiba band, making it narrower. Yu, Yu, Hu, Liu, and Wu (2011) replaced this heuristic search algorithm and utilized large margin criterion to have a better generalization ability on unseen test data. Although the R–K band usually provides better results than the optimized Sakoe-Chiba band, it still suffers from overfitting issues (Niennattrakul & Ratanamahatana, 2009) and in many cases the effectiveness of constrained and unconstrained DTW is the same (Wang et al., 2013). Moreover, Kurbalija, Radovanovic, Geler, and Ivanovic (2014) empirically proved that DTW is very sensitive to the introduction of global constraints and even a small change in an already narrow band changes relation between the time series. Last but not least, application of global constraint can make DTW rigid for real-time applications where there is no chance to do proper preprocessing and/or compensate the initial/ending shifts due to time or hardware limit. On the other hand, it has to be underlined that constraining is a must when fast DTW computation is prioritized, as the most effecting lower bounding functions and indexing methods are based on constraining (Keogh & Ratanamahatana, 2005; Lemire, 2009).

Locally penalized warping can also be used to avoid unwanted warpings (Sun, Lui, & Yau, 2006). In this case penalty is added to every non-diagonal movement of the warping path, i.e. the warping path constrains itself eventually. Instead of using a constant penalty, Clifford et al. (2009) suggested a penalty vector in which each non-diagonal step can have a different cost. A similar approach was presented by Taylor, Zhou, Rouphail, and Porter (2015) who applied the penalty as a scaling factor to the calculated cost. To provide a true data-adaptive constraint, Juhász (2007) used

[2] Review of DTW is omitted for brevity, a comprehensive discussion on it was written by Rabiner & Juang (1993).

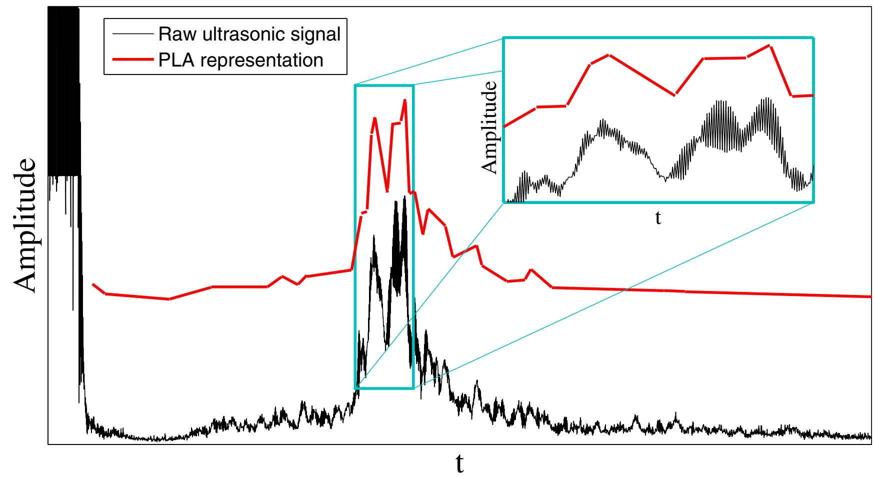
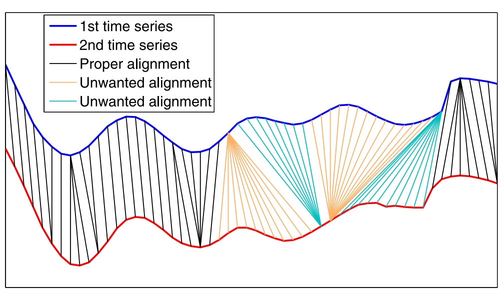
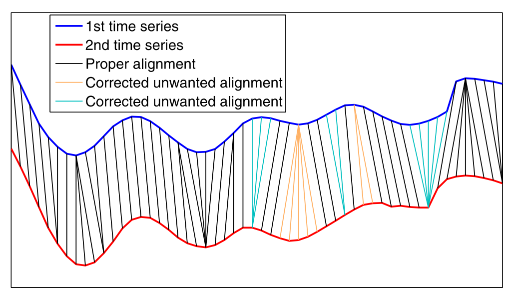


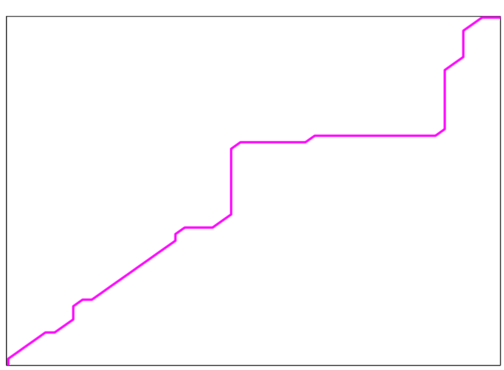
Fig. 1. Raw ultrasonic signal (black curve) and its piecewise linear approximation (PLA) (red lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



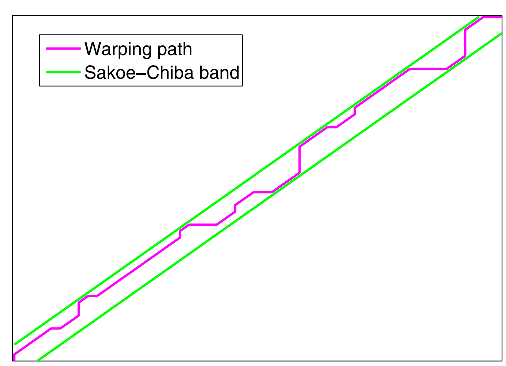
(a) Alignment generated by unconstrained DTW



(b) Alignment generated by the Sakeo-Chiba band constrained DTW



(c) Warping path of unconstrained DTW



(d) Warping path of the Sakeo-Chiba band constrained DTW

Fig. 2. Unwanted, pathological alignments (orange and turquoise lines in Figure (a)) created by unconstrained DTW and their correction using the Sakeo-Chiba band (green lines in Figure (d)) as global constraint. Sakeo-Chiba band constrains the warping path (purple line), which cannot stray far from the diagonal and thus the chance of pathological alignments is lowered. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

an adaptive weighting algorithm. Although all of these approaches give better control on the warping path, this additional control is also the drawback of penalized DTW. To select the best form of penalized DTW, deep and exhaustive knowledge of the underlying processes is necessary. The same is true for local constraints that also locally limit the warping path and their final form is usually based on heuristics as well (Rabiner & Juang, 1993).

Instead of artificially constraining the warping path, a feature (or feature set) representing the data points more suitably for warping can also be selected. Keogh and Pazzani (2001) were the first to propose such an approach, when they compared the estimated derivatives of each data point – i.e. the local trend – with the local dissimilarity measure of DTW. Usefulness of this method, called derivative dynamic time warping, was confirmed by several other researchers from chemistry (Gins, Van den Kerkhof, & Van Impe, 2012) to medicine (He, Wang, Mobley, Richman, & Grizzle, 2011) and several new representations and dissimilarity measures were introduced (Gullo et al., 2009; Li, Guo, & Qiu, 2011; Jeong, Jeong, & Omitaomu, 2011; Nakamura, Taki, Nomiya, Seki, & Uehara, 2012) that incorporated a similar, trend-based feature –

usually the slope of a section — to avoid pathological warpings. Zhang, Tang, and Duan (2015) moved one step further and utilized the shape context (Belongie, Malik, & Puzicha, 2002) of each data point as feature.

However, two separate studies suggested that considering one feature only does not resolve the strong data dependency of these methods. Xie and Wiltgen (2010) showed that regardless of the actual value of a data point or its derivative is used, unwanted warpings are unavoidable. To address this problem, they proposed a method in which both local and global trends of each data point are incorporated. Nonetheless, the separation of local and global trends makes the extension of this method impossible for any time series representation as the goal of the segmentation is to lose local information, i.e. to compact time series. The authors also showed through empirical tests using PLA segmented time series datasets that neither the value nor the slope-based (i.e. derivative) approach can outperform the other, as it can be seen in Fig. 4. Instead, the approach should be selected based on the dataset in question (Bankó & Abonyi, 2009).

Summarizing the aforementioned, it can be concluded that although considerable effort was put into finding an optimal constraining method, it has not been studied whether the warping path can be constrained directly by the local dissimilarity measure. This was the motivation to examine this hypothesis during verification of the introduced novel dissimilarity measure, called mixed dissimilarity measure for PLA (MDPLA). MDPLA is formulated as the combination of the mean-based dissimilarity measure (Keogh & Pazzani, 1999) and the authors' recently introduced trend-based approach (Bankó & Abonyi, 2009). Such combination makes MDPLA an ideal candidate for comparison with the underlying measures from the aspect of pathological warping. Moreover, MDPLA provides several advantages to the practicing engineer. It can be optimized for a given database, i.e. whether the focus has to be put on the mean of a PLA segment or it is more favorable to consider the trends. MDPLA also improves precision of the underlying measures even without resegmentation irrespectively of whether DTW is applied or not. Finally, as it will be shown, depending on the dataset, MDPLA makes the application of time warping unnecessary without sacrificing the quality of comparison.

The rest of the paper is organized as follows. In the next section, segmentation basics are shortly reviewed, in particular, piecewise linear approximation. Subsequently, the proposed MDPLA is presented and discussed why and how MDPLA helps to avoid unwanted alignments. Section 5 conducts a detailed empirical comparison with other time series representation/dissimilarity measure pairs. Next, it is discussed how MDPLA shortens the warping path and thus avoids pathological warpings, and how it is possible that non-warped MDPLA can outperform standard local dissimilarity measures even if DTW is applied. Finally, in Section 6 conclusions are made.

3. Theoretical background

To provide compact, comparable representation of time series, usually segmentation is applied where a number (or a set of numbers) represents each segment according to the selected representation method. PLA replaces the original data with not equally sized segments of straight lines, i.e. a PLA segment of time series x is described by four numbers:

$$\bar{x}(i) = [l_i, r_i, x(l_i), x(r_i)], \tag{1}$$

where l_i and r_i are the first and last (left and right) time coordinates of the ith PLA segment of x, $x(l_i)$ and $x(r_i)$ are the values of x in l_ith and r_ith time coordinates.

The segmentation problem can be framed in several ways (Fu, 2011), but its main goal is always the same: finding homogeneous segments by the definition of a cost function. This segmentation cost can usually be minimized by dynamic programming, which is computationally intractable for many real datasets (Himberg, Korpiaho, Mannila, Tikanmaki, & Toivonen, 2001). Consequently, heuristic optimization techniques such as greedy top-down or bottom-up techniques are frequently used to find good but suboptimal segmentation. Keogh, Chu, Hart, and Pazzani (2001) examined these heuristic optimization techniques through the application of PLA in detail and concluded if real-time (on-line) segmentation is not required, the best result can be reached by bottom-up segmentation.

Finding the optimal dissimilarity measure for any representation is a difficult task; however, the results of most dissimilarity measures can usually be improved by the application of DTW. DTW allows free selection of local dissimilarity measure, i.e. any dissimilarity measure or combination of dissimilarity measures can be used to compare segments. Gullo et al. (2009) examined several time series representations and compared their classification and clustering capabilities using DTW.

Best to our knowledge, Vullings, Verhaegen, and Verbruggen (1997) were the first to use PLA based DTW on ECG data, while Keogh and Pazzani (1999) gave a generalized dissimilarity measure between two PLA segments $\bar{x}(i)$ and $\bar{x}(j)$ for DTW by comparing the means of segments:

$$d_{MPLA}(\bar{x}(i), \bar{x}(j)) = \left(\frac{x(l_i) + x(r_i)}{2} - \frac{x(l_j) + x(r_j)}{2}\right)^2 \tag{2}$$

4. Mixed dissimilarity measure for piecewise linear approximation

The authors previously presented a new correlation-based segmentation method (Bankó & Abonyi, 2012) that based on the two homogeneity measures of principal component analysis (PCA). Using the Q reconstruction error — one of the homogeneity measures —, a multivariate time series can be segmented according to the change of the correlation structure among variables. The reconstruction error determines the hyperplanes in each segment, while the goal of the segmentation is to minimize the sum of the Euclidean distances between the original and the reconstructed variables in each segment such as for PLA. PLA also minimizes the Euclidean distances between the original data points and their reconstructed pairs (the closest points on the PLA segment). This distance (reconstruction error) for a PLA segment is shown in Fig. 3.

Considering that all of the PCA-based similarity measures[3] compare linear hyperplanes (principal components) of multivariate time series, it was desirable to generalize this approach to PLA. A PLA segment (i.e. a straight line) can be considered as the first (and only) principal component of the original data points. Thus, each PLA segment can be represented with the angle of its slope — i.e. its trend — and it can be used for comparison (Bankó & Abonyi, 2009). As PLA segments cannot be perpendicular to the time axis, no further restrictions are required and any angular-based dissimilarity measure can be used to compare the angles.

One significant difference between the PCA-based similarity measures is how they weight the angles between the hyperplanes. In the space of PLA segments, most of these measures can be reduced to the same equation for PLA segments $\bar{x}(i)$ and $\bar{x}(j)$:

[3] The dissimilarity (d) of two multivariate time series can be derived from an arbitrarily selected PCA-based similarity measure (s) ranged between 0 and k, i.e $d = k - s$.

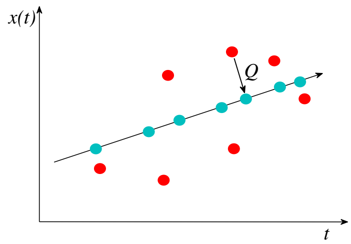


Fig. 3. Time series x (red dots) and its PLA representation (turquoise dots) with the Q reconstruction error. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$$d(\bar{x}(i), \bar{x}(j)) = 1 - \left| \cos\left(\operatorname{atan}\left(\frac{\tan(\bar{x}(i)) - \tan(\bar{x}(j))}{1 + \tan(\bar{x}(i))\tan(\bar{x}(j))} \right) \right) \right| \tag{3}$$

Eq. (3) would be a perfect dissimilarity measure for PCA, where the orientation of the hyperplanes is meaningless. For PLA segments, however, this information cannot be neglected. For example, if $\bar{x}(i) = 89^\circ$ and $\bar{y}(j) = 91^\circ$ are compared as hyperplanes, they are almost the same (the difference is only 2°); however, they are completely different from the univariate time series point of view as 89° belongs to an increasing trend while 91° belongs to a decreasing trend. Thus, the dissimilarity measure that compares the slopes of the PLA segments should measure the difference between the trends. For the sake of simplicity, the absolute difference was chosen:

$$d_{SPLA}(\bar{x}(i), \bar{x}(j)) = \left| \operatorname{atan}\left(\frac{x(r_i) - x(l_i)}{r_i - l_i} \right) - \operatorname{atan}\left(\frac{x(r_j) - x(l_j)}{r_j - l_j} \right) \right| \tag{4}$$

where $\bar{x}(i)$ and $\bar{x}(j)$ are the ith and jth PLA segment of univariate time series x, l_i and r_i are the first and last (left and right) time coordinates of $\bar{x}(i)$, $x(l_i)$ and $x(r_i)$ are the values of x in the l_ith and the r_ith time coordinates.

While the slope-based approach has definitely improved the results for some datasets, this was not always the case. Fig. 4 shows the results of the mean (MPLA) and the slope-based (SPLA) dissimilarity measures using the 1-NN classification test of the UCR time series classification/clustering repository (Keogh et al., 2011). It is also interesting to see that the difference between MPLA and SPLA is significant only for three datasets when DTW was not considered. When DTW was applied, however, seven datasets showed severe sensitivity to the change in local distance. This indicates that DTW is not able to properly align the data points and the alignment quality is highly dependent on the applied local dissimilarity measure (i.e. the considered feature) and on the dataset itself.

To demonstrate this dependency, two z-normalized time series were PLA segmented and compared with DTW using the above mentioned local dissimilarity measures. The z-normalized time series alongside with their PLA segmentation and the alignments created by DTW utilizing d_{MPLA} and d_{SPLA} as local dissimilarity measures are shown in Fig. 5(a) and (b).

Although both measures aligned most of the segments according to the expectations, neither of them did its job perfectly. d_{MPLA} was not able to align the plateaus, while d_{SPLA} did not align the increasing trends at the beginning properly. This example indicates that a mixture of the slope and the mean-based local dissimilarity measures is desired as it will decrease the chance of pathological alignments and eventually, it will provide better results. Thus, mixed dissimilarity measure d_{MDPLA} was defined by the authors between univariate time series x and y:

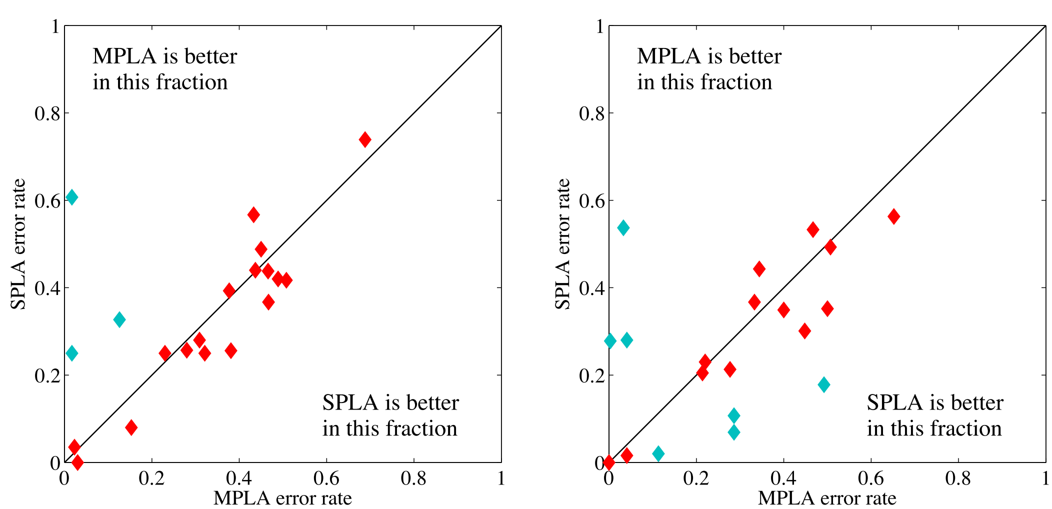
$$d_{\mathrm{MDPLA}}(x, y) = w_{\mathrm{MPLA}} d_{\mathrm{MPLA}}(x, y) + w_{\mathrm{SPLA}} d_{\mathrm{SPLA}}(x, y), \tag{5}$$

where w_{MPLA} and w_{SPLA} are the weights associated to the mean and the slope-based approaches.

In Fig. 5(c), time series is compared again, this time using d_{MDPLA} as local dissimilarity measure. As it can be seen, with proper selection of features and their weights, unintuitive alignments could be avoided without application of any constraint.

5. Validation results

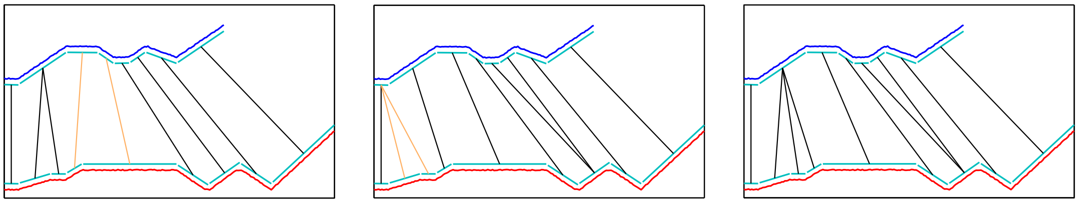
According to Keogh and Kasetty (2003), a collection of openly available and widely used datasets was selected for validation of MDPLA, namely the aforementioned datasets of the UCR time series classification/clustering repository (Keogh et al., 2011). The suggested 1-NN classification test was executed for PLA segmented time series using dissimilarity measures d_{MPLA}, d_{SPLA} and d_{MDPLA}. Being one of the most competitive representations, symbolic aggregate approximation (SAX) (Lin et al., 2003) was also included in the validation. Each representation/dissimilarity measure pair



(a) No warping, segments with corresponding indexes are compared

(b) DTW is utilized to create warping

Fig. 4. 1-NN test error rates of the UCR datasets using the slope (SPLA) or the mean (MPLA) of a PLA segment for comparison. Turquoise marks denote the databases that showed severe sensitivity to the change of the local dissimilarity measure.



(a) d_{MPLA} is used as local dissimilarity measure

(b) d_{SPLA} is used as local dissimilarity measure

(c) d_{MDPLA} is used as local dissimilarity measure

Fig. 5. PLA representations (turquoise lines) of two time series (red and black curves) compared with DTW using d_{MPLA}, d_{SPLA} and d_{MDPLA} as local dissimilarity measures. Black and orange lines denote the expected and the pathological warpings respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

was tested with and without DTW. Finally, to have a deeper understanding of MDPLA, precision-recall curves were also computed.

Before the results are discussed, the applied parameters and methods are exposed in detail to help reproduction. Each dataset is divided into a test and a train set by default. Using the latter ones, the number of segments was determined for every dissimilarity measure/representation pair as well the MDPLA weights (w_{MPLA} and w_{SPLA}) and the alphabet size for SAX. MDPLA weights were selected to satisfy $0 \leqslant w_{\text{MPLA}} \leqslant 1$ and $w_{\text{SPLA}} = 1 - w_{\text{MPLA}}$. For the training of SAX parameters, the same method was used as proposed by Lin, Keogh, Wei, and Lonardi (2007), i.e. segment number was searched from 2 up to half the length of the time series, with the segment number doubled in every subsequent iteration. Alphabet numbers were searched between 3 and 10 and in case of a tie the one with minimal segment number was selected.

The bottom-up segmentation algorithm was utilized to train the segment number for PLA-based measures. The time series of each training dataset was segmented into 5 to 30 segments. In case of multiple minimums, the lower number of segments was chosen. For MDPLA, the error rate can also be minimal for different combinations of w_{MPLA} and w_{MPLA}, in such a case the smaller w_{MPLA} was selected.

The classification algorithm was executed with and without utilizing DTW. For DTW, the applied constraints on the warping path, both global and local, also had to be defined. It was also assumed that no extensive knowledge exists on the databases, thus the most simple and yet widely used local constraint was selected, i.e. type I[4] of Rabiner and Juang (1993). Global constraint was not applied due to the issues mentioned at the beginning of this chapter and due to the fact that one motivation behind this validation was to show that warping path can be intuitively constrained by the selection of local dissimilarity measure only.

Results of the 1-NN test for all datasets and dissimilarity measures are presented in Table 1. Each cell contains the error rate for the given dataset/dissimilarity measure pair, the trained segment number, MDPLA weights (only for MDPLA) and SAX alphabet size (only for SAX). The best results (minimal error rates) both for warped and non-warped methods were marked by red color and the overall best result was typeset in bold. As it can be seen, MDPLA kept up with the expectations. It provided the lowest error rate for 8 and 14 of the 20 datasets in case of non-warped and warped methods, respectively. Displaying the results in Figs. 6 and 7, its performance increase is more evident over MPLA and SPLA.

To have a better overview on different approaches, precision-recall graphs were computed as well. For brevity, instead of showing all graphs, the average for each dissimilarity measure is presented in Fig. 8. As the number of elements belonging to a class differs, linear interpolation was applied. Note that, in this case linear interpolation was not used for the purpose to compute new precision-recall values. In such cases linear interpolation should be avoided as it creates an overly optimistic view on performance (Davis & Goadrich, 2006). Instead, all precision-recall values were computed – i.e. precision was computed for all possible values of true positives and thus all possible values for recall – and interpolation was used only for averaging the results. According to the expectations, DTW aided measures performed better than their non-warped versions while MDPLA clearly excelled all other methods.

5.1. Improving efficiency without resegmentation

One additional advantage of MDPLA is that it can improve the efficiency of MPLA and SPLA without modifying the segmentation itself, i.e. no resegmentation is necessary. This property is obviously beneficial when segment number cannot be changed, e.g., it is determined by a third party hardware or a software component. There is no argue that the best data for demonstrating this property should come from a real world application (production data, plant supervision data, real-time automotive sensor information, etc.); however, these kinds of data are rarely allowed to be published. Thus, UCR repository was utilized again. The number of segments was previously determined for each dataset in case of MPLA and SPLA, all there was left to do was to train w_{MPLA} and w_{SPLA} using the given segmentation. As previously, in case of a tie, the weights where w_{MPLA} is smaller was selected.

Figs. 9 and 10 show the improvement achieved over MPLA and SPLA. Considering the results, there are two things worth to mention: first, there is an improvement for almost all datasets and second, the change in results does not depend on the application of DTW.

5.2. Effect on computation power

Besides the additional effort of training the weights offline, MDPLA still requires more runtime than MPLA and SPLA. It is desirable to ask whether this runtime overhead worth it or not. In embedded applications, a widespread practice is to not utilize the whole processing power (and storage), instead, leave some space for future bug fixes, updates, etc. Thus, this non-used runtime will decide whether MDPLA can be used to replace MPLA or SPLA.

Another option to consider is switching warped MPLA or SPLA to non-warped MDPLA. In this case, the gain in runtime is obvious, DTW with even the tightest global constraint requires more processing power than non-warped MDPLA. Fig. 11 shows the results of such change. Warped MPLA can almost be freely substituted by non-warped MDPLA independently from the dataset, while SPLA require preliminary tests. It is also worth to mention that unwarped MDPLA provided better results in 19 cases from 40, which indicates that a properly selected feature or feature set –

[4] Symmetric, non-normalizable local constraint.

Table 1
Results of the UCR time series repository 1-NN classification algorithm. Cells contain the error rate for the given dataset/dissimilarity measure pair also the number of segments was trained using the train set. The trained alphabet size and w_{MPLA} are listed for SAX and MDPLA respectively. For each dataset, the best results (minimal error rates) were marked by red color, both with and without DTW, and the overall best result was denoted by bold letters.

Dataset	MPLA	SPLA	MDPLA	SAX	DTW on MPLA	DTW on SPLA	DTW on MDPLA	DTW on SAX
Synthetic Control	**.017** 28	.607 6	.023 24/.9	.020 16/10	.033 21	.537 5	.027 22/.7	.160 16/10
Gun-Point	.153 11	.080 20	.060 11/.1	.300 64/3	.113 11	**.020** 19	.027 8/.1	.440 4/7
CBF	.126 15	.327 6	.294 11/.4	.197 8/9	**.041** 24	.280 5	.064 12/.7	.051 64/10
Face all	.508 22	.417 9	.431 19/.8	.330 64/10	.448 30	.301 29	**.282** 28/.8	.348 64/10
OSU Leaf	.450 20	.488 17	.426 17/.1	.467 32/9	.492 17	178 30	**.165** 30/.1	.446 128/10
Swedish Leaf	.381 12	.256 11	.251 11/.1	.440 64/10	.277 30	.213 14	**.154** 28/.1	.725 8/10
50Words	.437 22	.440 10	.409 22/1	.334 128/9	.400 30	.349 27	**.279** 28/.1	.574 32/10
Trace	.030 8	**.000** 9	.010 7/.1	.460 128/10	**.000** 9	**.000** 8	**.000** 6/.1	.120 64/10
Two Patterns	.017 6	.250 5	.012 6/.8	.059 16/7	**.003** 11	.278 5	.004 8/.8	.004 16/8
Wafer	.023 25	.035 27	.018 21/1	**.003** 64/10	.041 21	.016 26	.014 28/.3	.074 8/9
Face four	.489 24	.420 13	.568 18/.4	**.307** 64/3	.500 19	.352 26	.364 19/.1	.364 128/9
Lightning-2	.377 9	.393 5	.311 9/.9	.377 16/5	.344 6	.443 19	.213 22/.8	**.180** 32/10
Lightning-7	.466 8	.438 6	.438 8/.5	**.370** 128/8	.507 9	.493 6	.411 9/.1	.384 16/10
ECG	.230 19	.250 5	.190 11/.3	**.100** 32/7	.220 19	.230 11	.190 11/.4	.250 8/9
Adiac	.688 12	.739 27	.642 13/.3	.900 32/10	.652 13	.563 30	**.458** 30/.1	.921 16/10
Yoga	.280 7	.257 7	.257 7/0	.235 128/8	.214 17	.205 19	**.150** 30/.1	.341 16/10
Fish	.309 12	.280 12	.166 7/.1	.474 128/10	.286 11	.069 28	**.040** 26/.1	.789 16/10
Beef	**.433** 10	.567 14	.467 6/.7	.600 128/8	.467 10	.533 14	**.433** 14/.5	.667 128/10
Coffee	.321 20	.250 8	.321 20/.3	.464 2/3	.286 28	**.107** 20	**.107** 20/0	.464 2/3
OliveOil	.467 14	.367 5	.300 30/.3	.833 2/3	.333 29	.367 5	**.267** 30/.3	.833 2/3

even such simple ones as for MDPLA — can outperform a general, one feature-based elastic measure, such as DTW coupled with the Euclidean distance.

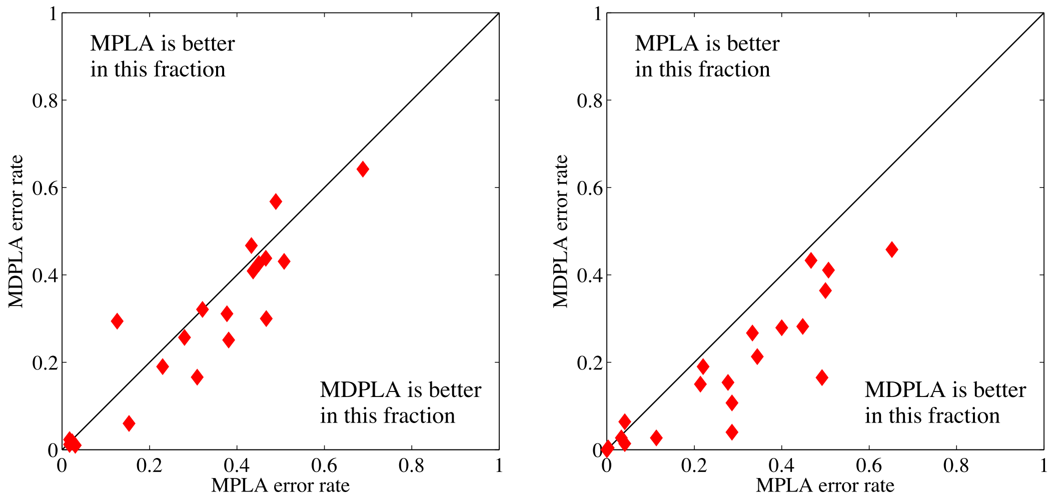
5.3. Avoiding unwanted warpings

Figs. 9 and 10 indicate that results were not degraded using DTW aided MDPLA for any of the 40 dataset/dissimilarity measure pairs. This is especially remarkable when it is compared to any other method that targets to make DTW better by eliminating pathological warpings. For example, application of R–K band (Ratanamahatana & Keogh, 2004) degrades the results for 6 of the 20 datasets compared to unconstrained DTW for the same 1-NN test (Keogh et al., 2011) using the original data. This definitely suggests that a properly selected local dissimilarity measure can prevent pathological warping as effective as any constraint.

In Section 4, it was demonstrated by a theoretical example how MDPLA can help to avoid unwanted alignments, now it is demonstrated empirically. The best way to determine whether a warping path is pathological or not is to inspect it manually, which would be intractable for the UCR datasets. However, considering the fact that all the previously reviewed constraints assume that each non-diagonal movement facilitates unwanted alignments and thus such movements must be minimized, it is possible to define *Tightness of Warping* for local dissimilarity measure d_m and dataset A:

$$ToW(A, d_m) = \frac{\sum_{i=1}^{\binom{|A|}{2}} l_i}{\sum_{i=1}^{\binom{|A|}{2}} max(m_i, n_i)}, \tag{6}$$

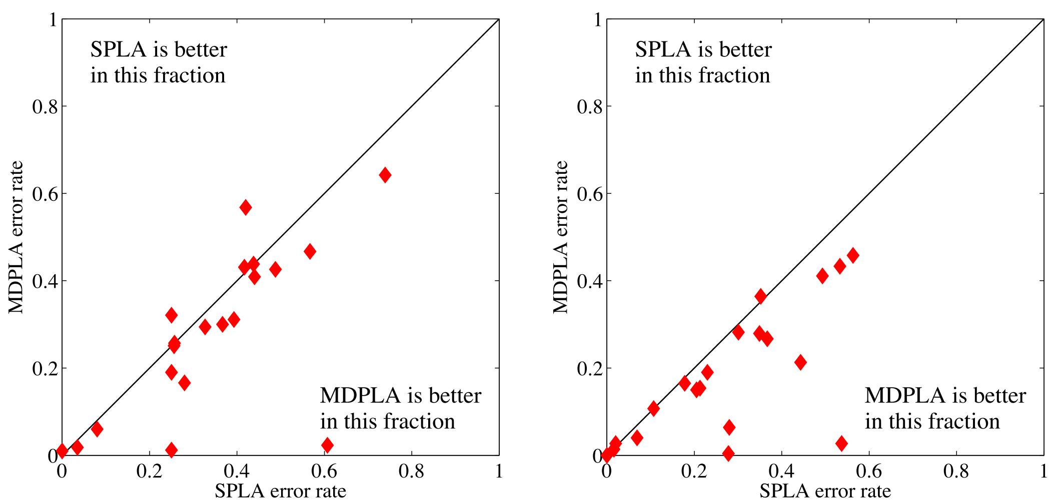
where $|A|$ denotes the number of time series in dataset A in which all time series are compared to each other with DTW using local dissimilarity measure d_m. Warping path lengths (l_i) resulted from these comparisons are summarized and divided by the sum of the minimum warping path lengths. Minimum warping path length for



(a) No warping, segments with corresponding indexes are compared

(b) DTW is utilized to create warping

Fig. 6. 1-NN test error rates of the UCR datasets using only the mean (MPLA) or both the mean and the slope (MDPLA) of a PLA segment for comparison.



(a) No warping, segments with the corresponding indexes are compared

(b) DTW is utilized to create warping

Fig. 7. 1-NN test error rates of UCR datasets using only the slope (SPLA) or both the mean and the slope (MDPLA) of a PLA segment for comparison.

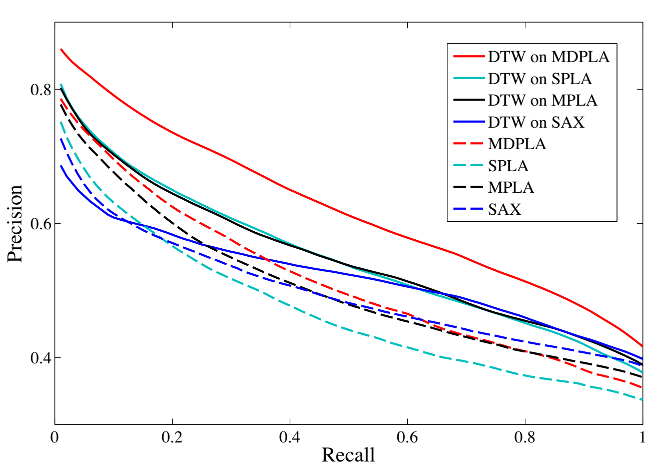
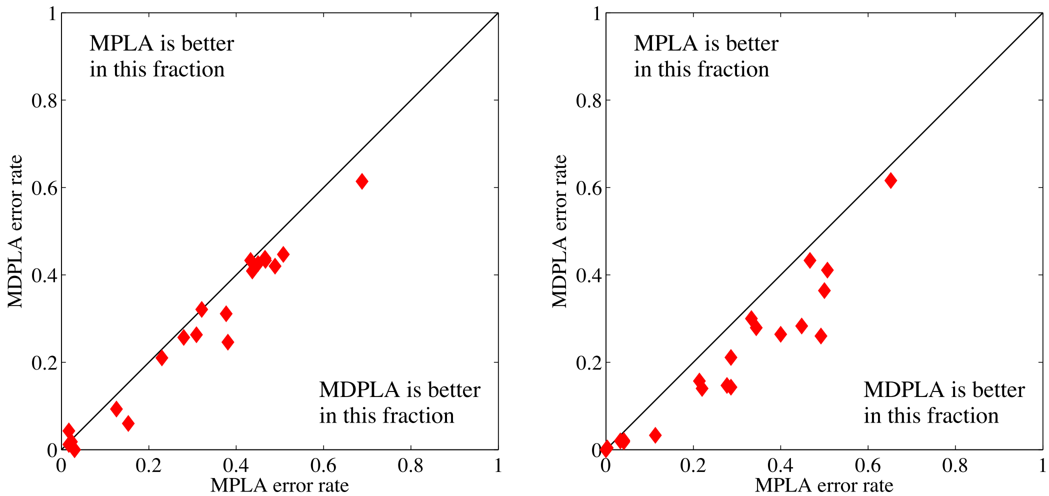


Fig. 8. Precision-recall graphs of the tested representations/dissimilarity measures.

the ith comparison equals the length of the diagonal of the warping matrix, i.e. the maximum of the lengths of time series used in the ith comparison (m_i, n_i).

$ToW(A, d_x) = 1$ means that all the warping path lies on the diagonal, while $ToW(A, d_y) = 3$ indicates that local dissimilarity measure d_y on the average triples the warping path in length and it most probably introduces pathological warpings.

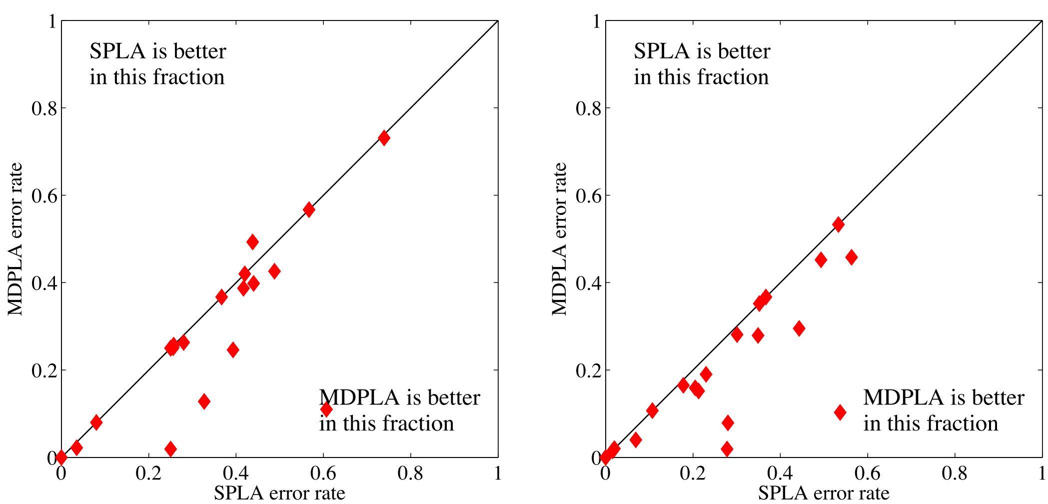
Using ToW, it is possible to compare local dissimilarity measures from warping point of view. All requirements are that the same constraints (local or global, penalties, etc.) and the same time series (or representation) shall be used. These requirements were fulfilled in Section 5.1 when MDPLA performance was assessed using the segmentations given by MPLA and SPLA. Table 2 contains the ToW values calculated using these segmentations. Warping path reduction was also calculated in percentage term, i.e. 100% and -100% would show when the warping path was reduced to the diagonal or when it was doubled, respectively. Looking at the



(a) No warping, segments the corresponding indexes are compared

(b) DTW is utilized to create warping

Fig. 9. Performance improvement of MDPLA when the segment number was originally trained for MPLA.



(a) No warping, segments with corresponding indexes are compared

(b) DTW is utilized to create warping

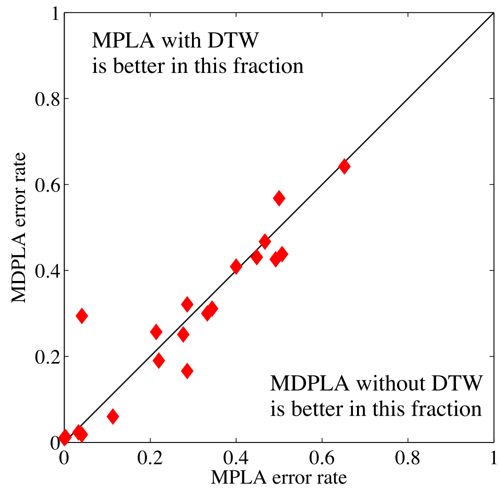
Fig. 10. Performance improvement of MDPLA when the segment number was originally trained for SPLA.

table, it can be seen that application of d_{MDPLA} as local dissimilarity measure really shortened the warping path in most cases — and what is also important — the warping path was never enlarged.

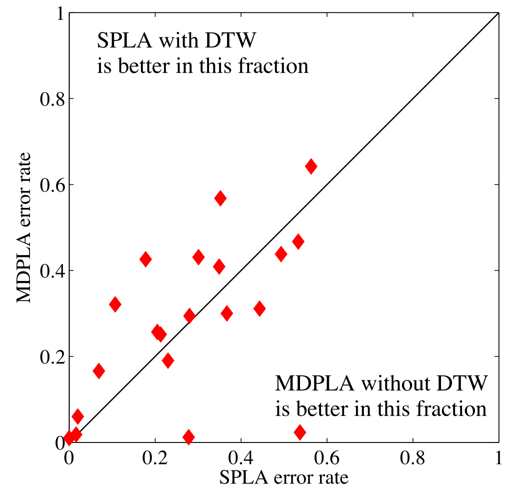
Comparing MDPLA to any other approach that targets pathological warpings, MDPLA has clear advantages. The most important is that no constraining is necessary, or to be more exact, the different features ensure that the warping path will be constrained automatically. This resolves three problems of global constraining methods: the necessity of synchronization of the staring and ending points, the general sensitivity of DTW to global constraints (Kurbalija et al., 2014) and the possibility of degraded accuracy. Another advantage of MDPLA compared to methods employing local constraints or penalty is that no exhaustive knowledge is required and the training phase for the local constraint or penalty can be omitted.

Similar approaches to MDPLA, such as DDTW (Keogh & Pazzani, 2001), SC-DTW (Zhang et al., 2015), etc., has been presented in the past in which a feature (or feature set) was used to represent the data point instead of its raw value. The main problem of these approaches is either that they focused on one feature only (and as it is can be seen in Fig. 4, it is often not adequate) or they require features derived from both local (i.e. the raw data) and global features (e.g. a segment), which makes them unusable in situations where only a given representation of the original time series (e.g. PLA segmented version) is available.

On the downside of PLA, it requires double the runtime as it compares two features instead of one. However, as it is discussed in the previous section, the increased accuracy arising from the fact that multiple features are considered and this often makes it possible to omit DTW and thus sparing run time at the end.



(a) Non-warped MDPLA compared to warped MPLA



(b) Non-warped MDPLA compared to warped SPLA

Fig. 11. Change in error rates when the warped MPLA and SPLA dissimilarity measures are replaced with non-warped MDPLA.

Table 2
Effect of the applied local dissimilarity measure on the length of warping path. Reduction shows how much the warping path was reduced (plus), or enlarged (minus) in percentage term. 100% would show that the warping path is reduced to its minimum, i.e. it always located on the diagonal.

Dataset	Segment number trained for MPLA			Segment number trained for SPLA		
	Tightness of Warping MPLA	Tightness of Warping MDPLA	Warping path reduction (%)	Tightness of Warping SPLA	Tightness of Warping MDPLA	Warping path reduction (%)
Syn Ctrl	1.439	1.199	54.68	1.176	1.066	62.31
Gun-Point	1.163	1.107	34.73	1.224	1.224	0.00
CBF	1.420	1.261	37.93	1.186	1.110	40.92
Face(all)	1.301	1.153	49.15	1.171	1.161	6.10
OSULeaf	1.334	1.142	57.63	1.205	1.196	4.10
SwedishLeaf	1.327	1.184	43.66	1.177	1.138	21.89
50Words	1.427	1.274	35.87	1.287	1.282	2.03
Trace	1.292	1.275	5.77	1.266	1.266	0.00
TwoPatterns	1.264	1.216	18.00	1.178	1.168	5.99
Wafer	1.368	1.335	8.82	1.412	1.350	15.07
Face (four)	1.277	1.140	49.50	1.163	1.163	0.00
lightning-2	1.221	1.155	29.87	1.309	1.265	14.15
lightning-7	1.300	1.227	24.32	1.221	1.169	23.31
ECG	1.296	1.193	34.85	1.165	1.150	8.79
Adiac	1.108	1.089	17.26	1.159	1.108	32.12
Yoga	1.266	1.200	24.69	1.228	1.222	2.61
Fish	1.125	1.117	6.34	1.213	1.168	21.31
Beef	1.329	1.169	48.57	1.152	1.152	0.00
Coffee	1.370	1.096	74.04	1.077	1.077	0.00
OliveOil	1.028	1.027	1.93	1.121	1.121	0.00

6. Conclusion

Finding a suitable time series representation alongside the matching dissimilarity measure has always been — and most probably always will be — a crucial task in time series data mining. While in many cases it is tempting to use DTW with any of the off-the-self distance measures created for a given representation, local dissimilarity measures considering one feature only often leave place for improvement.

The presented novel mixed dissimilarity measure for piecewise linear approximation (MDPLA) combines two features of a PLA segment, its mean and its trend. By combining these features and validating the resulted dissimilarity measure on the datasets of the UCR time series classification/clustering repository, it was possible to draw some valuable conclusions.

Combination of the features yielded to superior results over the cases when one of the features were considered only. In addition, MDPLA provides the advantage for practicing engineers who deal with PLA-based systems that this superiority can also be maintained without resegmentation.

One potential drawback of MDPLA is the increase in run-time. As it considers two features, it would obviously need double the run time to provide the improved results. However, this statement is not always valid. Glancing to Fig. 10, it can be concluded that either no significant difference can be seen between a DTW-aided measure that considers one feature only and the much faster computable, non-warped MDPLA or the difference has risen from the fact that a database favors one of the features irrespectively the application of DTW. The authors would like to underline this result, as although it is very tempting to use an off-the-self and popular comparison method like DTW or edit distances, often better results can be achieved — even both in run time and in accuracy — by properly selecting the feature (or feature set) to be compared.

Finally, it was also empirically demonstrated throughout the example of MDPLA that a dissimilarity measure combining several features helps avoiding pathological alignments by shortening the warping path of DTW. When the local distance of DTW considers multiple features for the comparison, pathological alignments can occur only if all of the features enables the alignment.

Acknowledgement

The contribution of János Abonyi was supported by the European Union and the State of Hungary, co-financed by the European Social Fund in the framework of TÁMOP 4.2.4. A/2–11-1–2012-0001, "National Excellence Program". The authors also thank Prof. Eamonn J. Keogh for providing us databases of the UCR time series repository as well András Róka and Jessica Lin who verified our SAX implementation.

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